13/01/2026

# Tutorial — 2.

Q. A mobile service provider wants to classify SMS message as Spam or Not Spam using a simple neural network (Perceptron)

$x_1$ = Specific Words and Phrases

$x_2$ = Links and Attachments

$x_3$ = Bias

| $x_1$ | $x_2$ | $x_3$ | Target (y) |
|---|---|---|---|
| 4 | 2 | 1 | 1 |
| 1 | 0 | 1 | 0 |
| 6 | 3 | 1 | 1 |
| 2 | 1 | 1 | 0 |

Solution:-

Perceptron (ML model)

1) Inputs : $x_1$ & $x_2$

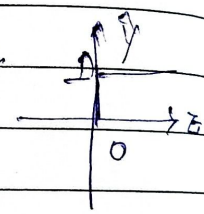
2) Weights : $w_1$ & $w_2$

3) Bias (b)

4) Weighted Sum (z) $z = x_1 w_1 + x_2 w_2 + b = \sum_{i=1}^{2} x_i w_i + b$

5) Activation function:- (Here: step function).

$$\hat{y} = \begin{cases} 0 ; & z < 0 \\ 1 ; & z \geq 0 \end{cases}.$$




6) Updated Weights

$$\boxed{w_i^{New} = w_i^{old} + \eta(y - \hat{y})x_i}$$

$\eta$ = learning rate

$y$ = Actual or True Target

$\hat{y}$ = Predicted Target

(i) $x_1 = 4$, $x_2 = 2$, $b = 1$

Let $w_1 = 0$, $w_2 = 0$

$z = w_1 x_1 + w_2 x_2 + b = 1.$ $\hat{y} = 1.$

Prediction is correct.

Iteration = Epoch

Quick Work

Page No.:

Date:

M T W T F S S

Q2. An email service provider wants to classify incoming emails as Spam or Not Spam using a single network model. The training dataset is shown in below:

| $x_1$ | $x_2$ | $x_3$ | Target (y) |
|---|---|---|---|
| 5 | 3 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 7 | 5 | 1 | 1 |
| 2 | 2 | 1 | 0 |
| 6 | 1 | 1 | 1 |

(i) $x_1 = 5$, $x_2 = 3$, $b = 1$

Let $w_1 = 0$, $w_2 = 0$. $\eta = 1$

$z = 1$. $\hat{y} = 1$ $y = 0$.

$w_1$ New $= 0 + 1(0-1) \times 5 = -5$

$w_2^{New} = 0 + 1(0-1) \times 3 = -3$.

Now

$w_1 = -5$, $w_2 = -3$, $x_1 = 1$, $x_2 = 0$, $b = 1$.

$z = -5 + 1 = -4$. $\hat{y} = 0$. $y = 0$.

Prediction is Correct

Q3 An e-commerce company predicts whether a customer will make a purchase base on time spent on website (min) and no. of items viewed.

| Time on site ($x_1$) | Items viewed ($x_2$) | Bias ($x_3$) | Target ($y$) |
|---|---|---|---|
| 2. | 3 | 1 | 0 |
| 15 | 25 | 1 | 1 |
| 8 | 12 | 1 | 1 |
| 3 | 5 | 1 | 0 |
| 12 | 20 | 1 | 1 |

Sol:-

(i) $x_1 = 2,\ x_2 = 3,\ b = 1,\ w_1 = 0,\ w_2 = 0.$

$z = 2\times0 + 3\times0 + 1$

$= 1. \quad \hat{y} = 1,\ y = 0. \quad \eta = 1$

$w_1^{new} = 0 + 1[0-1]\times 2. = -2.$

$w_2^{new} = 0 + 1(0-1)\times 3 = -3.$

Now,

$w_1 = -2,\ w_2 = -3,\ x_1 = 15,\ x_2 = 25,\ b = 1$

$z = -2\times15 + -3\times25 + 1 = -104.$

$\hat{y} = 0,\ y = 1.$

$w_1^{new} = -2 + 1[1-0]\times 15 = 13.$

$w_2^{new} = -3 + 1(1-0)\times 25 = 22.$

Now

$w_1 = 13,\ w_2 = 22,\ x_1 = 8,\ x_2 = 12,\ b = 1$

$z = 13\times8 + 22\times12 + 1 = 369$

$\hat{y} = 1,\ y = 1$

Prediction is correct.